

Pipes, redirection, xargs

Shell power often means connecting small tools

Pipe, redirect, and feed arguments

- The vertical bar connects stdout of the left command to stdin of the right
- Greater-than overwrites a file with stdout
- Xargs reads items from stdin and builds argument lists, handy when you have many filenames
- Together they turn a noisy log into a short list of errors

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Pipes, filters & redirection

Compose pipelines and redirects on a sample design log. See each stage's stdout and files written by > / tee.

Starter example: cat sim.log | grep ERROR | tee errors.txt | wc -l — count ERROR lines while saving a copy.

Load starter example

Challenges 0 / 22 cleared

Grep ERROR: Show only ERROR lines from sim.log.

Show hintCheckNextIdle

Grep ERRORCount ERROR linesRedirect to filetee + countGrep WARNINFO onlyLine countFirst 3 lines

Last 3 linesScoreboard linesDriver errorsFind timeoutFind finishingNon-INFO headFirst two ERRORS

tee filtered logxargs check filesxargs itemSorted uniqueBackground jobKill a jobUnexpected X

Recipe shortcuts

Click to fill the pipeline box (still Run + Check for challenges).

cat sim.log | grep ERRORcat sim.log | grep ERROR > errors.txtcat sim.log | grep ERROR | tee errors.txt | wc -l

echo src/main.v src/alu.v | xargs -n 1 echo checkcat sim.log | grep -v INFO | head -n 5cat sim.log | grep WARN

cat sim.log | grep -v ERROR | grep -v WARNcat sim.log | wc -lcat sim.log | head -n 3cat sim.log | tail -n 3

Real shell practice

```
wsl - module11-pipes/examples/pipes

# Track A - real Unix

# real shell session (Track A)

$ head -5 sample.log
[INFO] Module 3 pipes example - sample log for grep practice
[INFO] Start simulation at 2024-02-03 10:00:00
[INFO] Loading design from src/top.v
[WARNING] Unused signal 'debug_q' in block top
[INFO] Compilation complete

$ grep ERROR sample.log
[ERROR] Assertion failed at line 42: expected 0x0, got 0x1
[ERROR] Mismatch in output file
[ERROR] 2 tests failed, 1 passed

$ grep -i warning sample.log
[WARNING] Unused signal 'debug_q' in block top
[WARNING] Timeout threshold approaching
[WARNING] Clock period may be too short for timing

$ grep -c ERROR sample.log
3

$ cat sample.log | grep INFO | head -5
[INFO] Module 3 pipes example - sample log for grep practice
[INFO] Start simulation at 2024-02-03 10:00:00
[INFO] Loading design from src/top.v
[INFO] Compilation complete
[INFO] Running testbench

$ grep -i error sample.log | wc -l
3
```

Real shell — pipes and grep

Real shell practice — try these

```
# head -5 sample.log — peek at the start of the log  
head -5 sample.log
```

```
# grep ERROR sample.log — lines containing ERROR  
grep ERROR sample.log
```

```
# grep -i warning sample.log — case-insensitive warning lines  
grep -i warning sample.log
```

```
# grep -c ERROR sample.log — count matching lines  
grep -c ERROR sample.log
```

```
# cat sample.log | grep INFO | head -5 — pipe: filter then take first five  
cat sample.log | grep INFO | head -5
```

Pitfalls to watch

- A pipe only carries stdout by default
- Overwrite with one greater-than destroys the previous file
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few pipeline challenges after the starter
- On the real shell
- When you are ready, take the short quiz, then continue to sort, uniq, and cut